

COURSE PROJECT PROPOSAL CSE23602 – MACHINE LEARNING

PROJECT TITLE	Cattle Disease Prediction Using Machine Learning			
DOMAIN OR FIELD	Finance			
TEAM MEMBERS	1GA23CS024	1GA23CS027	1GA23CS037	
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I. INTRODUCTION TO DOMAIN AND PROBLEM IDENTIFIED	The domain of this project lies in the field of financial technology, commonly referred to as FinTech, combined with advanced techniques from machine learning. Financial technology has transformed the way individuals manage their money by enabling digital platforms for investing, saving, and financial planning. Despite these advancements, a significant gap still exists in providing intelligent, personalized, and easily understandable financial guidance to users. Most available platforms either offer generalized advice that does not consider individual differences or present complex analytical data that is difficult for non-experts to interpret. The problem identified in this project is the lack of accessible and explainable financial advisory systems that can adapt to individual user profiles. Many individuals are unable to determine their risk tolerance, choose appropriate investment strategies, or plan their finances effectively due to limited financial literacy and absence of personalized guidance. Existing systems often operate as black boxes, providing recommendations without explaining the reasoning behind them. This creates a lack of trust and limits their usability. To address this issue, the proposed solution is an AI-powered financial advisor system that utilizes machine learning to analyze user data and generate personalized investment strategies. The system is designed to predict a user's risk profile, recommend an optimal asset allocation, and provide clear explanations for its suggestions. By integrating intelligent decision-making with transparency, the project aims to bridge the gap between complex financial analysis and user-friendly financial planning tools.
II. BACKGROUND	Financial advisory services have traditionally been delivered by human experts who analyze a client's financial situation and provide tailored investment recommendations. While effective, such services are often expensive and accessible only to high-net-worth individuals. With the

	advancement of technology, automated advisory platforms known as robo-advisors have emerged, such as Betterment and Wealthfront. These platforms use predefined algorithms to automate investment decisions and make financial services more accessible. However, these systems still face several limitations. They often rely on static rules and do not fully adapt to individual user behavior or changing financial conditions. Additionally, they lack transparency, making it difficult for users to understand how recommendations are generated. This lack of explainability reduces user confidence and limits the effectiveness of such systems. The integration of machine learning into financial advisory systems has the potential to overcome these limitations. Machine learning models can analyze large volumes of financial and behavioral data to identify patterns and make predictions. Techniques from Machine Learning enable the system to learn user preferences and risk tolerance, while concepts such as Portfolio Optimization allow for the construction of efficient investment portfolios. Additionally, Natural Language Processing can be used to generate explanations and interact with users in a more intuitive manner. These advancements form the foundation for developing intelligent and user-centric financial advisory systems.
III. OBJECTIVES	The primary goal of this research project is to bridge the technological gap in rural veterinary diagnostics by achieving the following specific objectives: To design a personalized investment recommendation system: The project focuses on generating tailored portfolio suggestions by considering individual financial conditions, goals, and risk tolerance instead of relying on generic investment strategies. To implement portfolio optimization techniques: The system applies concepts from Portfolio Optimization to construct an optimal allocation of assets that balances expected returns with associated risks. To incorporate explainable AI for transparency: The objective is to ensure that every recommendation is supported by clear and interpretable explanations, enabling users to understand the reasoning behind the system's decisions. To develop a user-friendly interface: The project aims to provide an intuitive platform where users can input their financial information and easily visualize recommendations and insights.

	• To enable goal-based financial planning: The system is intended to align investment strategies with specific user-defined goals such as retirement planning, savings growth, or major expenditures.
IV. METHODOLOGY	The methodology adopted for this project follows a structured pipeline that integrates data processing, machine learning modeling, financial analysis, and system deployment to build an intelligent financial advisory system. The process begins with data collection, where relevant financial and user-specific data are gathered from reliable sources. Historical market data such as stock prices, indices, and interest rates are obtained from platforms like Yahoo Finance, while user-related data including income, expenses, savings, and financial goals are collected through structured inputs or simulated datasets. This combination of market and user data forms the foundation for building a personalized advisory system. Once the data is collected, preprocessing is performed to ensure its quality and usability. This step involves handling missing values, removing inconsistencies, normalizing numerical features, and performing feature engineering to derive meaningful attributes such as savings ratio, income stability, and expenditure patterns. Proper preprocessing is essential to improve the performance and reliability of machine learning models. The next stage involves the development of a risk profiling model. In this step, classification algorithms from Machine Learning such as logistic regression and random forest are used to categorize users into different risk levels, including conservative, moderate, and aggressive. The model is trained on historical or simulated data and evaluated using performance metrics such as accuracy and F1-score to ensure its effectiveness. Finally, the system is deployed as a full-stack application. The backend is developed using frameworks such as FastAPI or Flask to handle data processing and model inference, while the frontend is built using modern web technologies to create an interactive dashboard. This allows users to input their financial details, view recommended portfolios, and understand the reasoning behind the system's suggestions in a user-friendly manner.
V. REFERENCE	[1] Markowitz, H. (1952). <i>Portfolio Selection</i>. [2] Lundberg, S. M., & Lee, S. I. (2017). <i>A Unified Approach to Interpreting Model Predictions</i>. [3] Heaton, J., Polson, N., & Witte, J. (2017). <i>Deep Learning in Finance</i>.

	[4] Sirignano, J., & Cont, R. (2019). <i>Universal Features of Price Formation in Financial Markets: Perspectives from Deep Learning</i>. [5] Gu, S., Kelly, B., & Xiu, D. (2020). <i>Empirical Asset Pricing via Machine Learning</i>. [6] Fischer, T., & Krauss, C. (2018). <i>Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions</i>. [7] Khandani, A. E., Kim, A. J., & Lo, A. W. (2010). <i>Consumer Credit-Risk Models via Machine-Learning Algorithms</i>. [8] Chen, T., & Guestrin, C. (2016). <i>XGBoost: A Scalable Tree Boosting System</i>.